

Bio-Inspired and Deterministic Pathfinding Algorithms for Microrobot Navigation in Vascular Networks: A Comparative Study

Francisco Marques | *Instituto Superior Técnico, Universidade de Lisboa*

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Abstract

It is known that medicine has been evolving towards personalized and more targeted treatments and methods [1]. This includes swarms of drug carrier bots that can deliver drugs to specific places in the body using the vascular system as pathway. Therefore, these bots must have efficient and effective navigation algorithms behind. This work investigates and compares deterministic algorithms (A*) [4] vs bio-inspired ones (ACO) [5] all adapted to the most important vascular system parameters such as width, direction of the flow and distance. They were both tested in graphs of 50 nodes and 500 nodes. Results reveal a clear trade-off: A* is faster and deterministic but limited by its geometric heuristic, while ACO is slower but finds lower-cost biological pathways in larger networks. In a 500-node network, ACO achieved a lower path cost than A* (31.11 vs 32.94) without Poiseuille's law, while A* outperformed ACO under the Poiseuille model (82.83 vs 127.94), with A* consistently faster (Simple Model: 0.0007s vs 0.1366s; Poiseuille's Model: 0.0008s vs 0.0980s).

Introduction:

Being able to deliver personalized and targeted medicine to patients is a top priority for modern research and that includes nanotechnology, such as microrobots [1]. These bots could have the ability to deliver drugs to specific targets using the vascular system as pathway. This has an even more important role in cancer treatments, in which we could be able to deliver treatment with extreme precision, saving the patients from the extreme toxic systemic effects that these therapies may have. Currently these bots are already being studied using magnetic guidance systems [2,3]. These face significant limitations including pulsatile flow variations, trajectory errors [2] from time-multiplexed sequence parameters, and signal noise during tracking [2]. These constraints motivate the exploration of autonomous path-finding algorithms as a complementary approach.

For these microrobots to work properly it is critical that they can travel through the complex vascular system that can be seen as an extremely complex graph. Finding the optimal path in such graph is not trivial and, many times, the classical deterministic algorithms, while optimal, rely on geometric heuristics that may not capture the complexity of biological cost functions.

In this paper, the vascular network was modelled as directed graphs that have specific biological properties. It was modelled using 3 properties: diameter, length and direction of flux. The objective is to test two different algorithms to find the optimal path in a vascular network. A deterministic algorithm was used: A* [4]; and a bio-inspired one: Ant Colony Optimization (ACO). They were both

tested in a 50 and 500 nodes graph and to better reflect real vascular physics, edge costs were computed using Poiseuille's law, which models blood flow resistance as a function of vessel diameter and length, heavily penalizing narrow vessels.

Methods:

The vascular system was modeled using a directed graph (DiGraph) (Figure 1) where each node represents a vascular bifurcation and each edge represents a blood vessel. The edges have biological properties specific to the cardio-vascular system: diameter (mm), length (cm) and the direction of the blood flux. The cost of each path between edges was calculated using the Poiseuille law: $R = (8\eta L) / (\pi r^4)$ [6], where η is the blood viscosity (0.004 Pa.s), L is the length and r the radius of the vessel. This law extremely penalizes the narrow vessels – a reduction of 50% in diameter increases the resistance 16 times. When the direction of the flux is in the opposite direction than the intended, the cost is multiplied by a factor of 2.

Two algorithms were compared: the A* and the ACO. The A* is a deterministic algorithm for pathfinding, this means that it is always assured to find the best path. It uses the formula: $f(n) = g(n) + h(n)$, where $g(n)$ is the accumulated cost of the path and $h(n)$ is the Euclidean heuristic [4]. The ACO is a bio-inspired algorithm based on the behavior of ant colonies [5]. The idea is that each ant builds a probabilistic path based on pheromones. Applied to this project, each agent constructs a path probabilistically based on pheromone trails and vascular edge costs. The probability of choosing the next node is proportional to the pheromone level and the inverse of the edge cost. After each iteration, pheromone

levels evaporate globally and are reinforced on edges belonging to shorter paths (edges traversed more frequently accumulate more pheromone) biasing future agents towards lower-cost routes. (Parameters: $\alpha=1$, $\beta=2$, evaporation rate=0.5, 5 agents, 100 iterations)

Both algorithms were tested on two synthetically generated graphs with random coordinates (seed=42): 50 nodes and 500 nodes, with 2 to 5 edges per node. Each algorithm was executed from the start (node zero) to the final node. The metrics that were measured were: the path cost, the execution time and the number of nodes in the path. On the bigger graph the experiment was repeated with and without Poiseuille's law to evaluate its impact on the cost model.

Results & Discussion:

In small graphs (8 and 50 nodes) it was concluded that both algorithms reach the same path (Figure 2). This happens because the space of solutions is small enabling the ACO to find the same path of the A*, which is the best one.

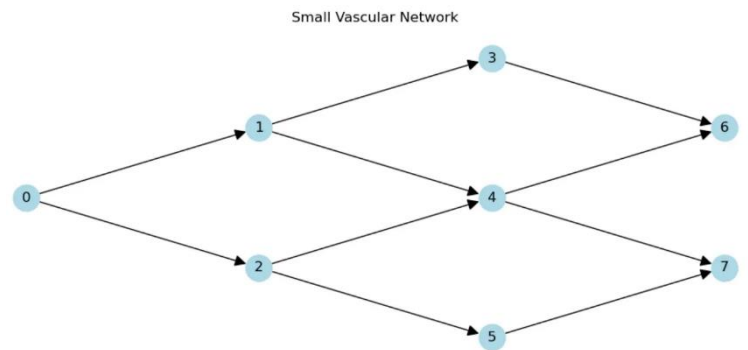


Figure 1- Small vascular network (8 nodes) modelled as a directed graph.

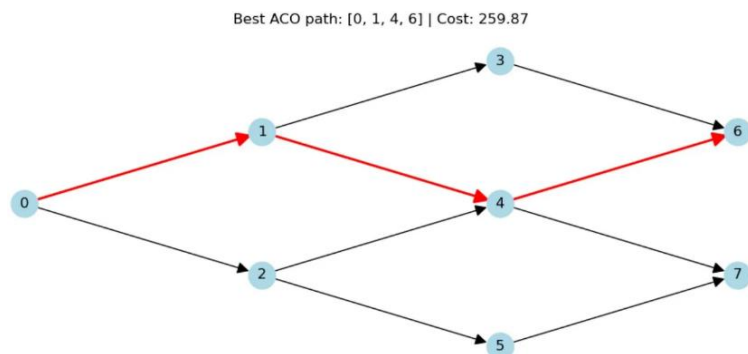


Figure 2- Best path found by ACO on the small network (path: [0, 1, 4, 6], cost: 259.87). Red edges indicate the optimal route.

Table 1- Comparison of A* and ACO algorithms across different graph sizes and cost models.

Graph	Cost Model	Algorithm	Path Cost	Time (s)	Nodes in Path
8 nodes	Poiseuille	A*	259.87	-	3
8 nodes	Poiseuille	ACO	259.87	-	3
50 nodes	Poiseuille	A*	32.46	0.0003	3
50 nodes	Poiseuille	ACO	32.46	0.0804	3
500 nodes	Simple	A*	32.94	0.0007	4
500 nodes	Simple	ACO	31.11	0.1366	6
500 nodes	Poiseuille	A*	82.83	0.0008	5
500 nodes	Poiseuille	ACO	127.94	0.0980	6

On the other hand, in large graphs (500 nodes) with the simple cost model ((length / diameter) $\times$ flux), the ACO finds a smaller cost path (31.11 vs 32.94) in comparison to the A*. This is since the ACO explores more alternative paths and because the A* heuristic guides it towards paths that are more geometrically direct but that don't necessarily have a smaller vascular cost. With this model the ACO converges after 2 iterations (Figure 3). On the same 500 nodes graph, but this time with Poiseuille's law as the cost model, the results change. This law ($R = (8\eta L) / (\pi r^4)$) strongly penalizes narrow vessels (that have smaller radius) and creates some much larger cost differences: 82.83 for the A* vs 127.94 for the ACO. Notably, under the Poiseuille model the A* optimal path increased from 4 to 5 nodes compared to the simple model. This reflects how Poiseuille's law dramatically alters edge costs — a path with fewer nodes but narrower vessels becomes more expensive than a longer path through wider ones, forcing the A* to recalculate a geometrically different route.

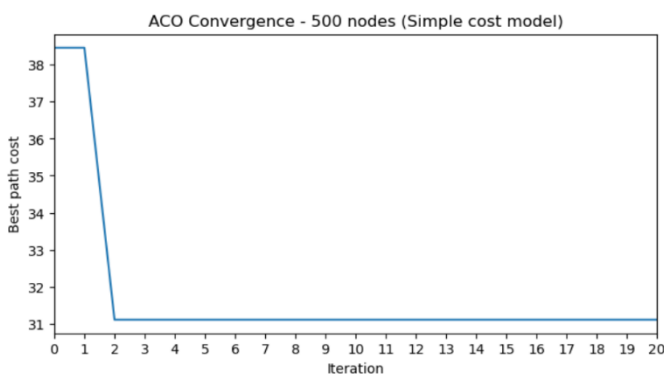


Figure 3- ACO convergence on 500-node network (Simple cost model). Best path cost stabilizes at ~31.11 after ~5 iterations.

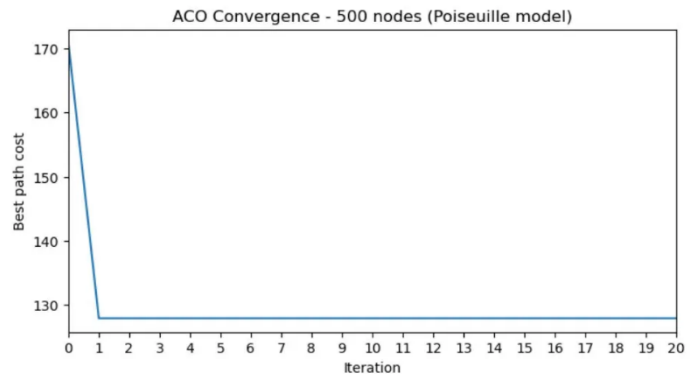


Figure 4- ACO convergence on 500-node network (Poiseuille model). Algorithm converges rapidly in 1 iteration, indicating premature convergence.

It is also important to note that with this model the ACO converges prematurely in a single iteration with this cost model (Figure 4). This means that the algorithm finds and stabilizes in one solution before exploring enough of the rest of the solution space. This happens with the Poiseuille's cost model because the costs are much bigger in magnitude (82-127 vs 31-32 in the simple model). This fact makes the relative differences between paths to be proportionally smaller, the first reasonable path that the bots find quickly dominates the pheromones, making the following iterations to only reinforce this path without exploring alternatives. In terms of time, the A* is consistently faster than the ACO: 0.0003s vs 0.0804s (50 nodes graph); 0.0007s vs 0.1366s (500 nodes graph – Simple); 0.0008s vs 0.0980s (500 nodes graph – Poiseuille). This can be explained by the fact that A* is deterministic and performs a single guided traversal of the graph, guaranteeing optimality without iterative exploration. The full comparison is summarized in Table 1.

During this project several limiting factors were found and could guide some future work. The graphs were all synthetic graphs which meant that they don't reflect the real vascular topology [2]. The flux factor was

either 1 or 2, this is a simplification that isn't based on physics [6]. The pheromone model in the ACO algorithm is a computational abstraction, meaning that if these nano-bots were physical there would not be real communication between them like ants do. It was only tested one pair of origin-destination for graph. The ACO parameters (alfa, beta, evaporation) were not optimized systematically.

For future work it would be possible to use real vascular databases such as VasculSynth and have a real flux model with differential pressure. The ACO parameters could be optimized with grid search, and multiple origin-destination pairs could be tested. It would also enrich the paper a comparison with other bio-inspired algorithms such a graph-adapted version of PSO (Particle Swarm Optimization). Future work could also explore realistic inter-agent communication models, including signal latency and limited range, to evaluate their impact on swarm convergence in vascular environments. Additionally, implementing a 2D or 3D visualization would make the work more realistic and less abstract.

References:

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